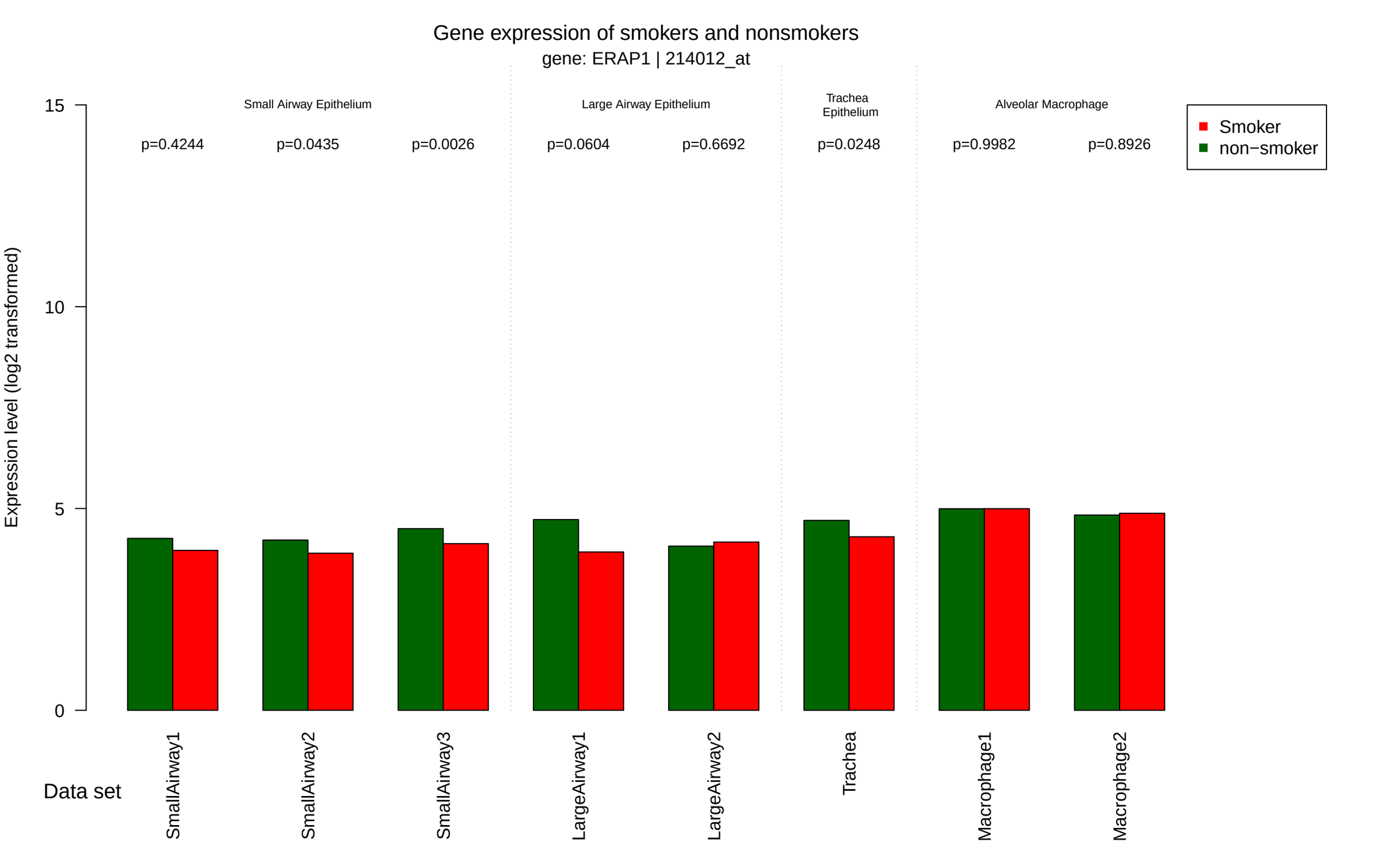
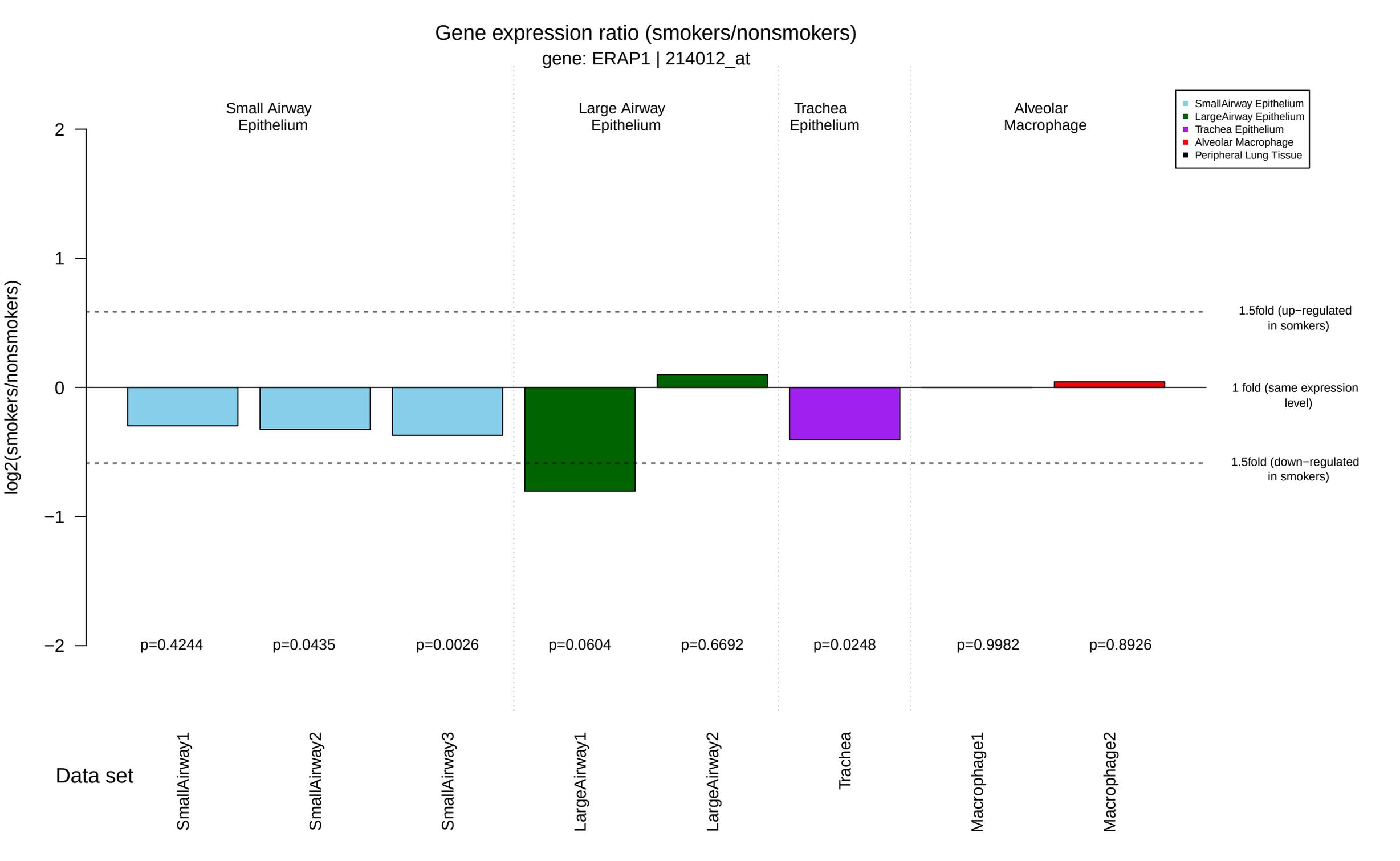
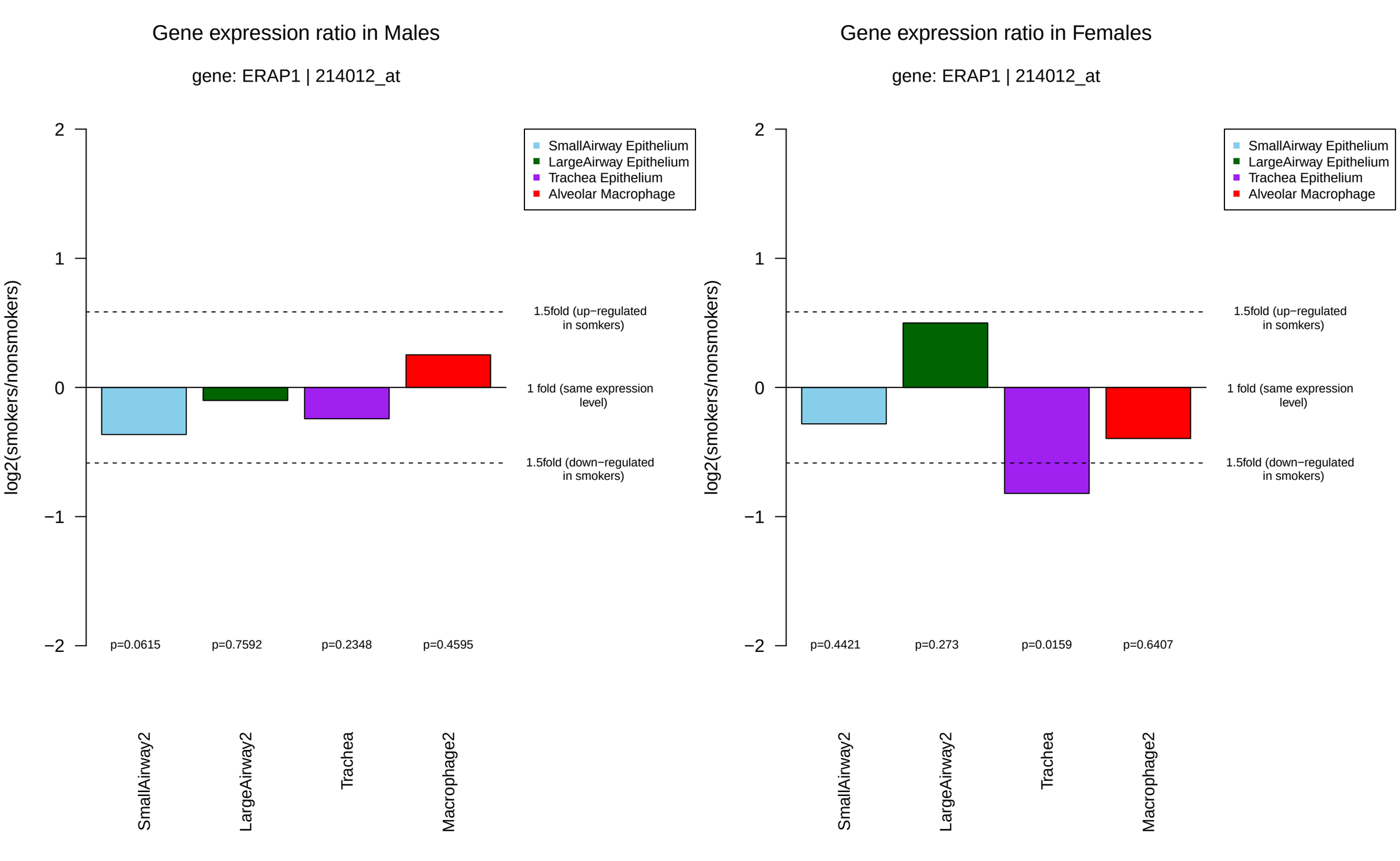
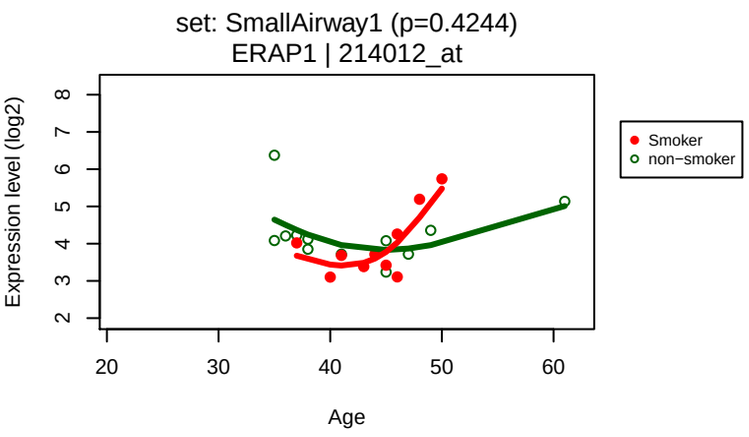
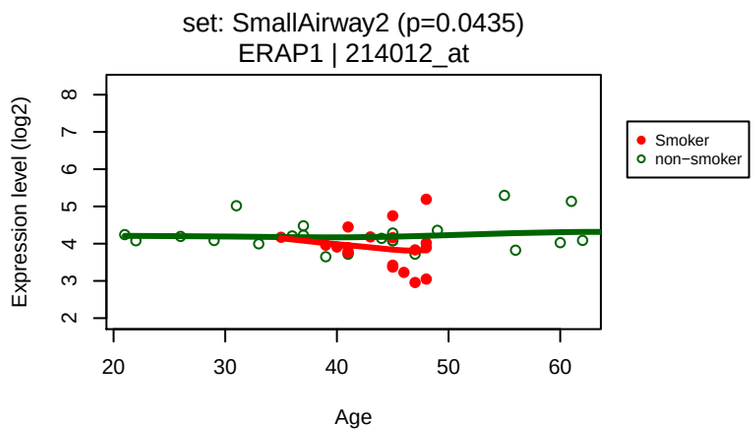
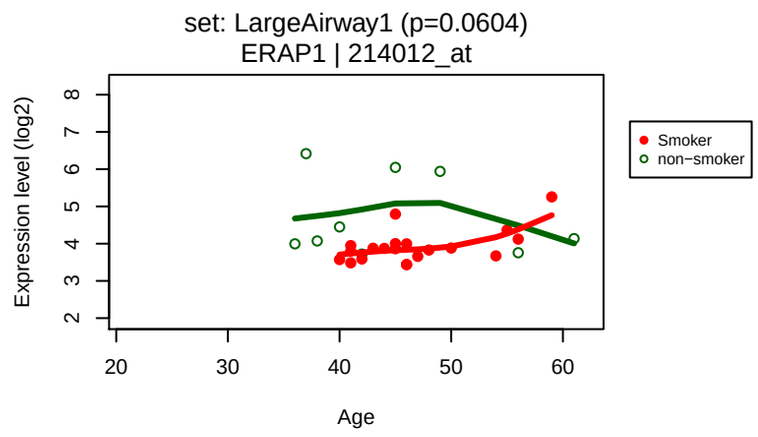

Gene expression of smokers and nonsmokers
gene: ERAP1 | 214012_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4244
p=0.0435
p=0.0026
p=0.0604
p=0.6692
p=0.0248
p=0.9982
p=0.8926
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

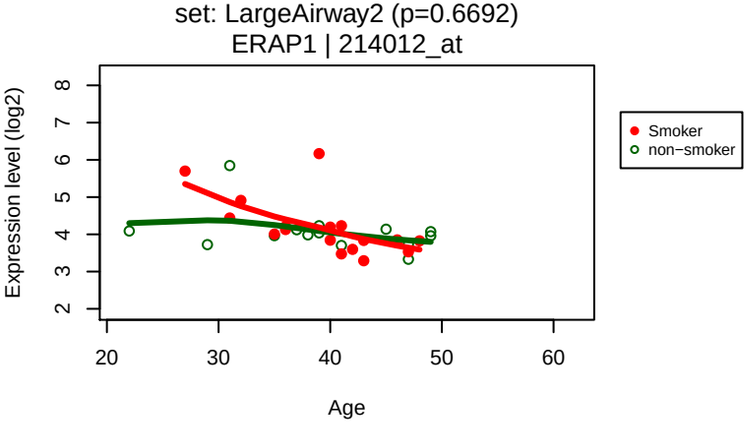

Gene expression ratio (smokers/nonsmokers)
gene: ERAP1 | 214012_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4244
p=0.0435
p=0.0026
p=0.0604
p=0.6692
p=0.0248
p=0.9982
p=0.8926
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

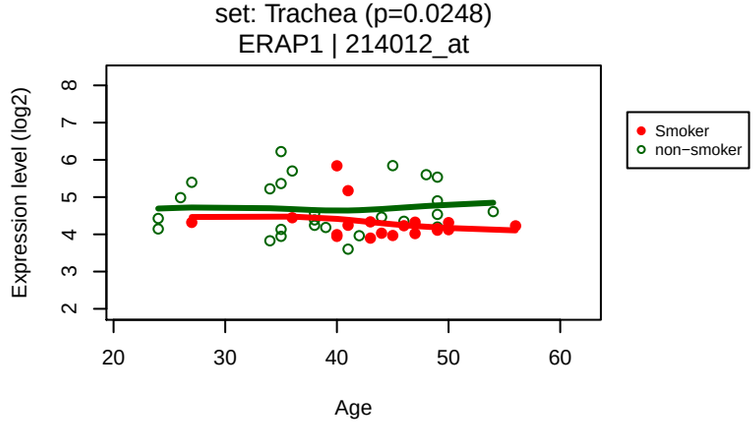

Gene expression ratio in Males
gene: ERAP1 | 214012_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0615
p=0.7592
p=0.2348
p=0.4595
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ERAP1 | 214012_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4421
p=0.273
p=0.0159
p=0.6407
SmallAirway2
LargeAirway2
Trachea
Macrophage2

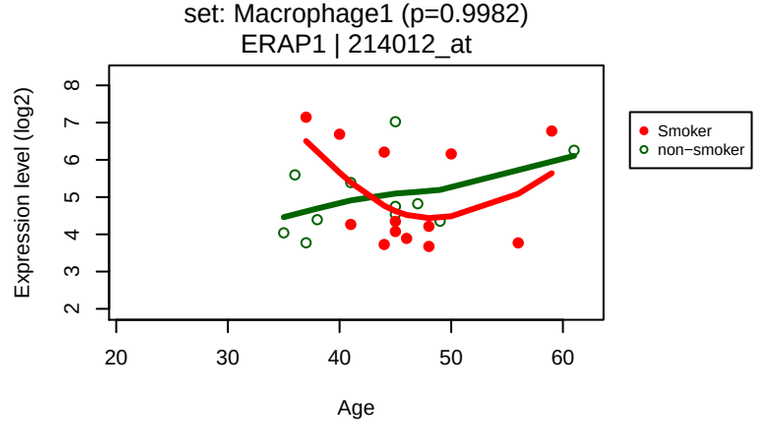

set: SmallAirway1 (p=0.4244)
ERAP1 | 214012_at
Expression level (log2)
Age
Smoker
non−smoker

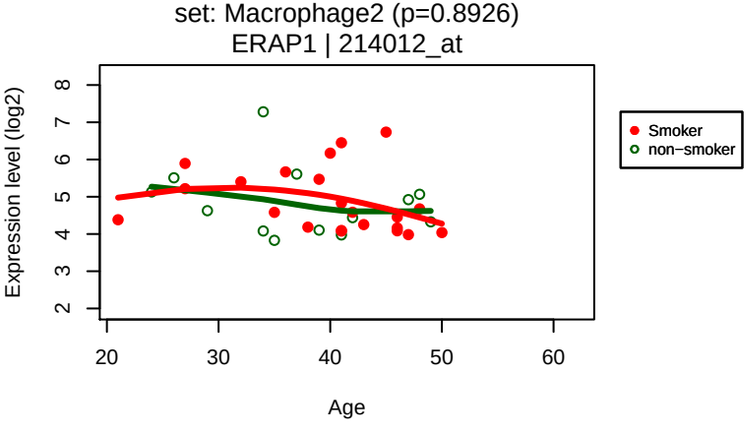

set: SmallAirway2 (p=0.0435)
ERAP1 | 214012_at
Expression level (log2)
Age
Smoker
non−smoker

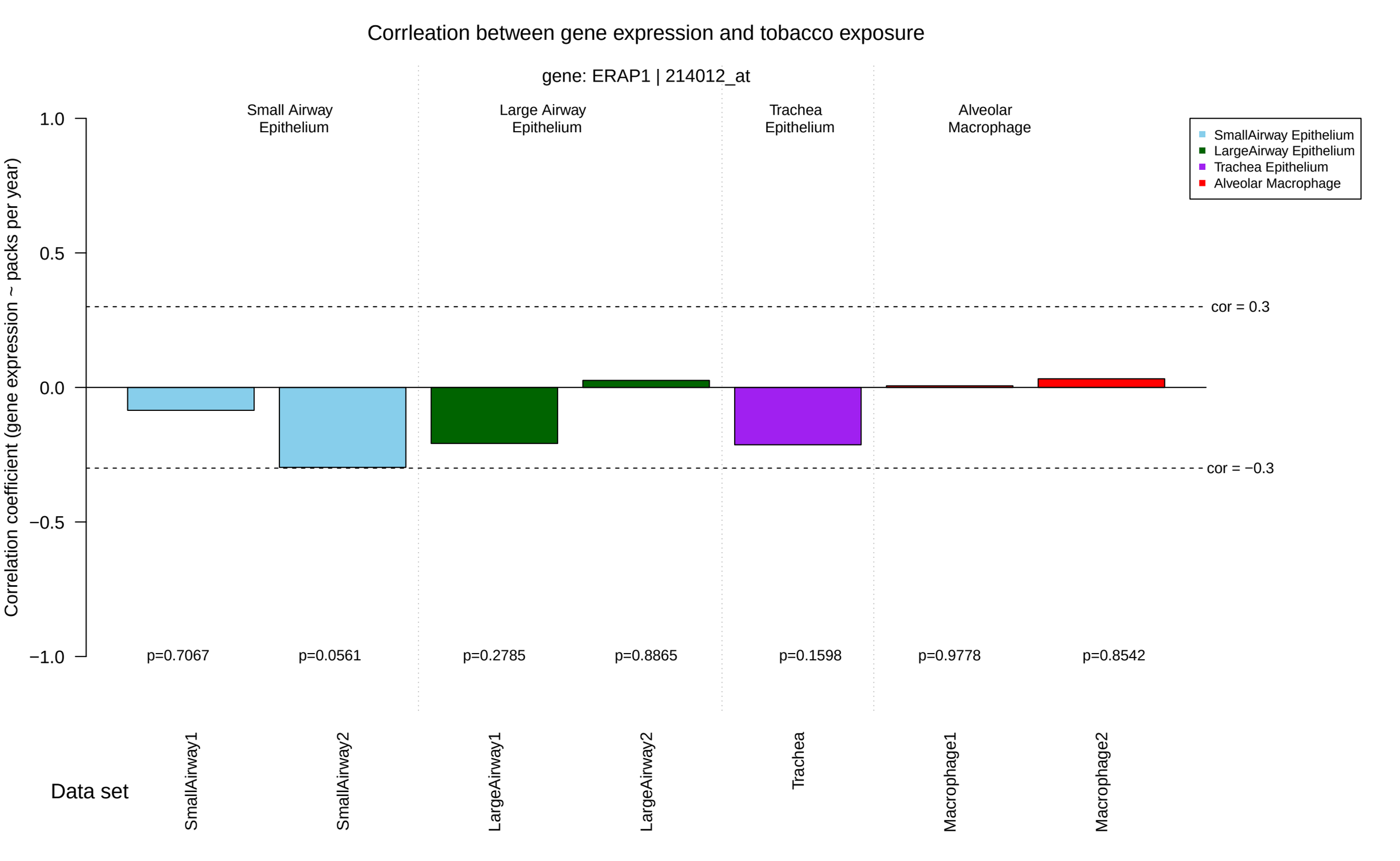

set: LargeAirway1 (p=0.0604)
ERAP1 | 214012_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6692)
ERAP1 | 214012_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0248)
ERAP1 | 214012_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9982)
ERAP1 | 214012_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8926)
ERAP1 | 214012_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ERAP1 | 214012_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7067
p=0.0561
p=0.2785
p=0.8865
p=0.1598
p=0.9778
p=0.8542
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2